Stock Price Forecasting Using Deep Learning Architectures

Technical Report

M. Ahsan

July 2026

Abstract

This report presents a comparative study of four deep learning architectures — Artificial Neural Network (ANN), Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Transformer — applied to next-day stock price forecasting. Using ten years of daily historical data across five stocks from different sectors (AAPL, ABT, JPM, XOM, WMT), the project covers exploratory data analysis, feature engineering, model development, hyperparameter tuning, and rigorous evaluation. Bonus extensions include an attention-augmented LSTM, walk-forward validation, comparison against traditional machine learning baselines, ten-day-ahead forecasting, and an interactive Streamlit dashboard. GRU emerged as the most consistent and reliable architecture, outperforming the other three models on four of five stocks.

Contents

1 Introduction

Stock price forecasting remains one of the most challenging applications of time-series modeling due to high noise, non-stationarity, and regime shifts in financial markets. This project implements and rigorously compares four deep learning architectures for next-day closing price prediction across five stocks spanning different sectors: technology (AAPL), healthcare (ABT), finance (JPM), energy (XOM), and retail (WMT). Beyond the core comparison, the project extends into several bonus areas, including an attention-augmented LSTM variant, walk-forward validation to simulate live trading conditions, comparison against traditional machine learning baselines, ten-day-ahead price forecasting, and an interactive dashboard for result exploration.

2 Data and Methodology

2.1 Data Collection

Ten years of daily OHLCV (Open, High, Low, Close, Volume) data were collected via the yFinance API for five stocks: AAPL, ABT, JPM, XOM, and WMT. Exploratory analysis confirmed no missing values or duplicate records across any of the five datasets.

2.2 Exploratory Data Analysis

EDA covered daily returns, rolling volatility, moving averages (20/50/200-day), trend decomposition, correlation structure, and outlier detection using the interquartile range (IQR) method. Figures 1 through 6 summarize these analyses.

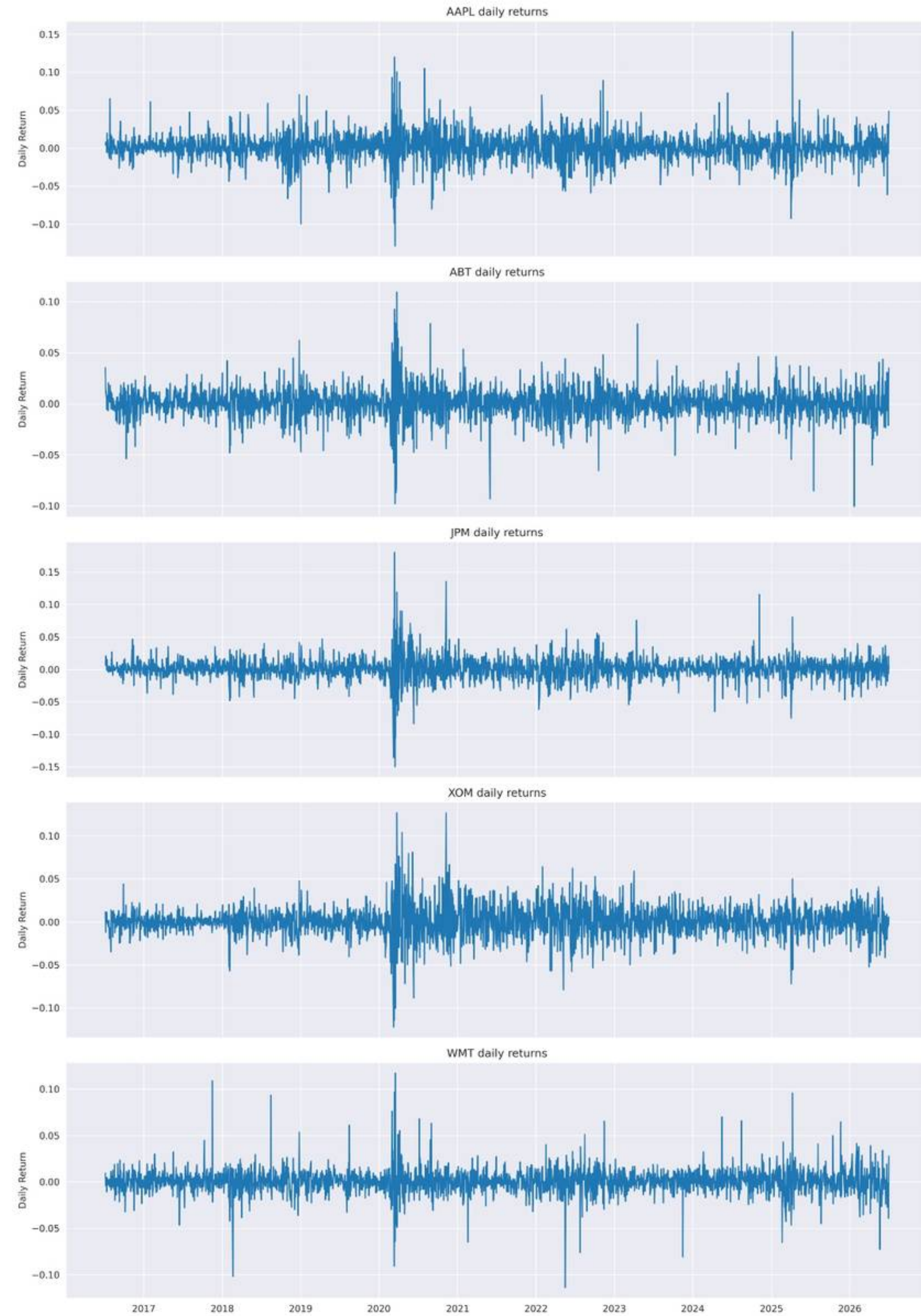

Figure 1: Daily closing price trends for all five stocks over the ten-year period.

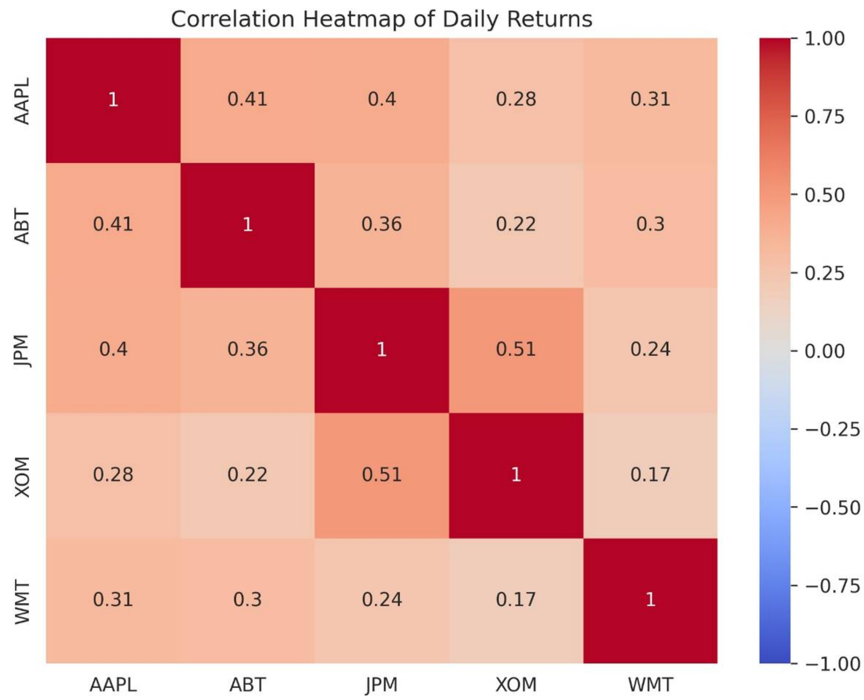

Figure 2: Feature correlation heatmap across engineered technical indicators.

Figure 3: 20-day, 50-day, and 200-day moving averages overlaid on closing price per stock.

Figure 4: Rolling 30-day volatility across all five stocks.

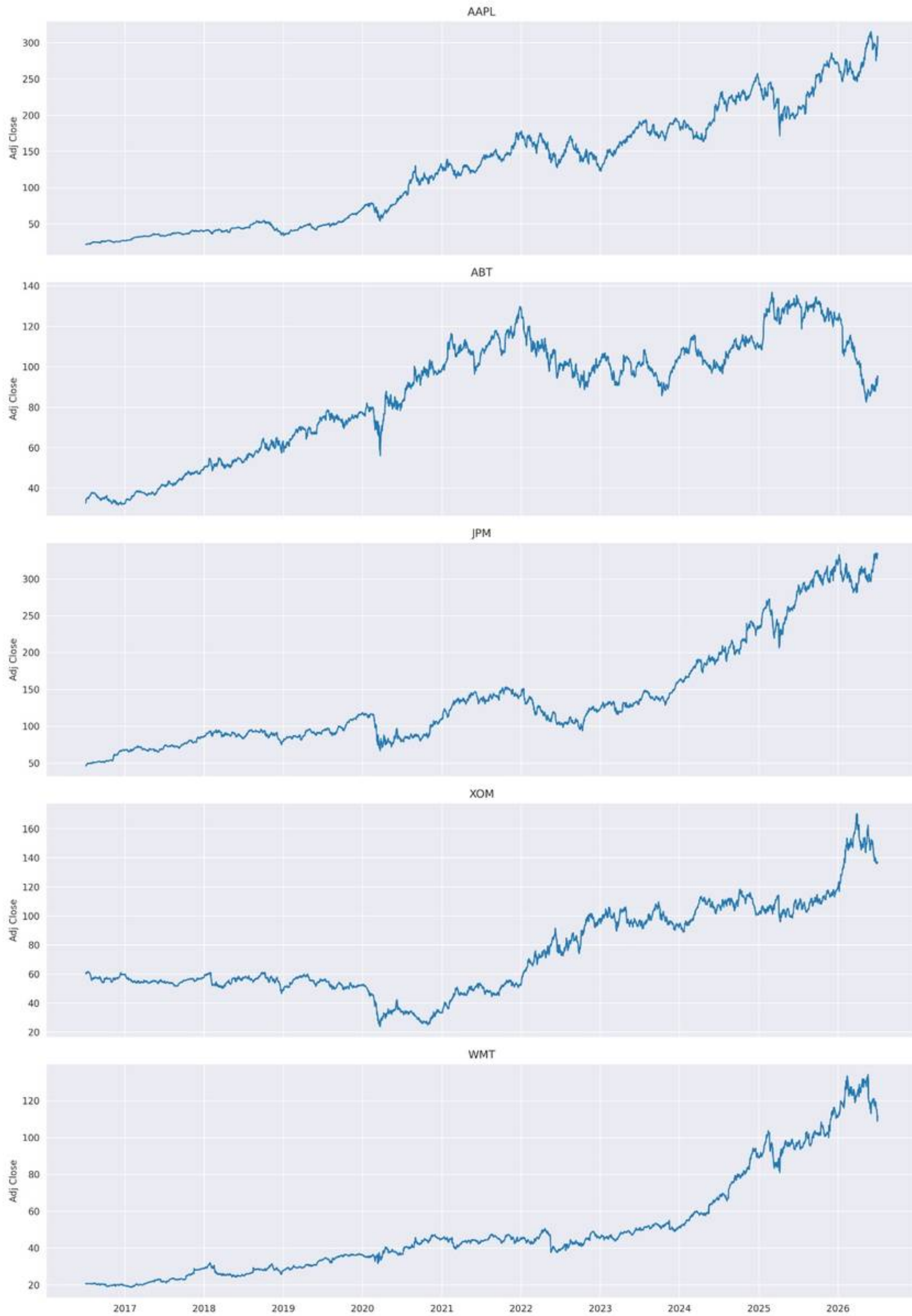

Figure 5: Trend decomposition (trend, seasonality, residual components) per stock.

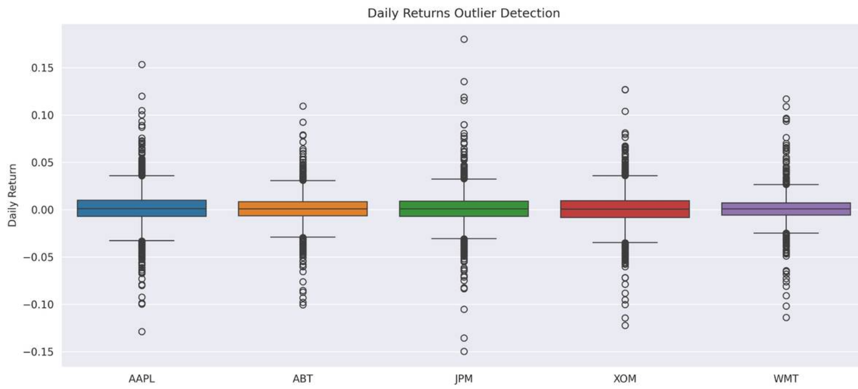

Figure 6: Outlier detection using the IQR method, highlighting anomalous price movements.

2.3 Feature Engineering and Preprocessing

Technical indicators including SMA, EMA, RSI, MACD, and Bollinger Bands were engineered alongside price returns, log returns, and rolling statistics (mean, standard deviation, min, max), expanding each dataset to 22 features. All features were normalized using Min-Max scaling. Data was split chronologically (training / validation / test) to avoid lookahead bias, and two sequence window sizes (30-day and 60-day) were evaluated, with the 60-day window producing superior results for the GRU architecture.

3 Model Development

Four architectures were implemented and trained with early stopping (patience of 10 epochs) to prevent overfitting:

- **ANN**: A feed-forward network using flattened sequence windows as a naive baseline, with no explicit temporal structure.
- **LSTM**: A multi-layer Long Short-Term Memory network with dropout regularization, processing sequential windows directly.
- **GRU**: A multi-layer Gated Recurrent Unit network, structurally simpler than LSTM with fewer gating mechanisms.
- **Transformer**: A multi-head self-attention block with feed-forward layers, designed to capture long-range dependencies.

4 Hyperparameter Tuning

Hyperparameter tuning was performed on AAPL using the GRU architecture, testing sequence length, batch size, dropout rate, and optimizer choice individually before combining the best values.

Table 1: Hyperparameter tuning results (AAPL, GRU).

Hyperparameter	**Best Value**	$\mathbf{R^2}$
Sequence length	window_60	0.9539
Batch size	16	0.9444
Dropout	0.2	0.9322
Optimizer	rmsprop	0.9628
Final combined config	window60_batch64_drop0.1_adam	0.9390

Interestingly, the final combined configuration did not exceed the best individual tuning results, suggesting some hyperparameter interactions offset one another — a useful insight for future tuning strategies, such as testing rmsprop specifically within the combined configuration.

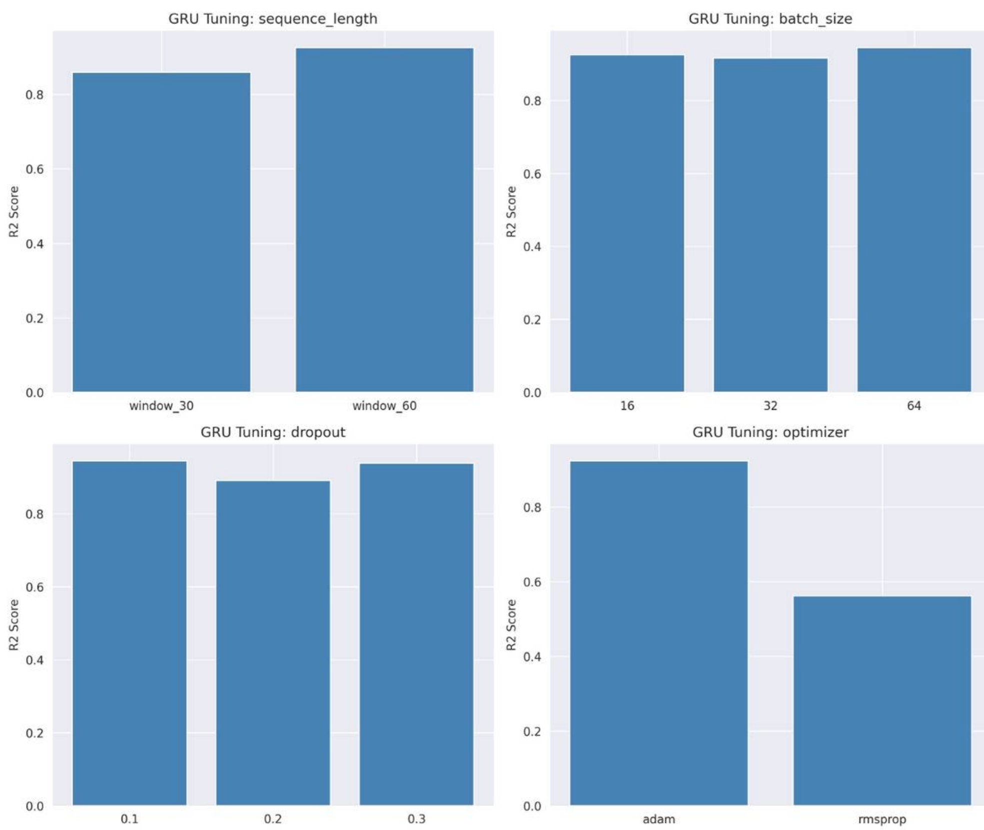

Figure 7: GRU hyperparameter sensitivity analysis across sequence length, batch size, dropout, and optimizer.

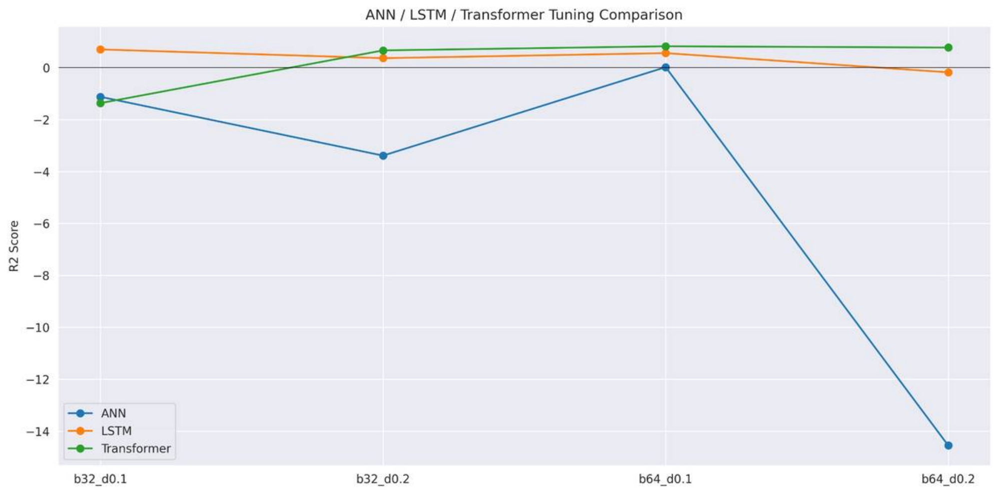

Figure 8: Hyperparameter sensitivity comparison for ANN, LSTM, and Transformer architectures.

5 Model Evaluation

All reported results are sourced from the validated benchmark file (`VALIDATED_final_results.csv`), covering all four architectures across all five stocks on the held-out test set.

Table 2: Validated test-set performance across all models and stocks.

Ticker	**Model**	**MAE**	**RMSE**	**MAPE (%)**	**R^2**	**Epochs**
AAPL	ANN	0.0393	0.0501	3.55	0.4175	11
AAPL	LSTM	0.0261	0.0306	2.28	0.7823	40
AAPL	Transformer	0.0426	0.0488	3.77	0.4481	30
AAPL	**GRU**	**0.0118**	**0.0154**	**1.06**	**0.9452**	32
ABT	ANN	0.0515	0.0573	5.44	0.6700	12
ABT	**LSTM**	**0.0135**	**0.0169**	**1.50**	**0.9713**	48
ABT	Transformer	0.0327	0.0402	3.77	0.8378	52
ABT	GRU	0.0129	0.0177	1.48	0.9684	25
JPM	ANN	0.1282	0.1376	8.03	-2.1824	13
JPM	LSTM	0.1501	0.1557	9.37	-3.0738	22
JPM	Transformer	0.0820	0.0948	5.21	-0.5123	36
JPM	**GRU**	**0.0529**	**0.0645**	**3.35**	**0.3008**	13
XOM	ANN	0.0559	0.0801	4.92	0.4379	11
XOM	LSTM	0.0298	0.0386	2.58	0.8698	14
XOM	Transformer	0.0273	0.0353	2.42	0.8907	100
XOM	**GRU**	**0.0150**	**0.0188**	**1.38**	**0.9690**	14
WMT	ANN	0.1584	0.1860	8.83	-1.6768	11
WMT	LSTM	0.3404	0.3502	19.19	-8.4940	49
WMT	Transformer	0.3419	0.3570	19.24	-8.8626	80
WMT	**GRU**	**0.1472**	**0.1643**	**8.17**	**-1.0891**	20

Table 3: Average performance per model across all five stocks.

Model	**Avg R^2**	**Avg MAPE (%)**	**Avg Epochs**
GRU	**0.4189**	**3.09**	20.8
ANN	-0.4667	6.15	11.6
Transformer	-1.4397	6.88	59.6
LSTM	-1.7889	6.99	34.6

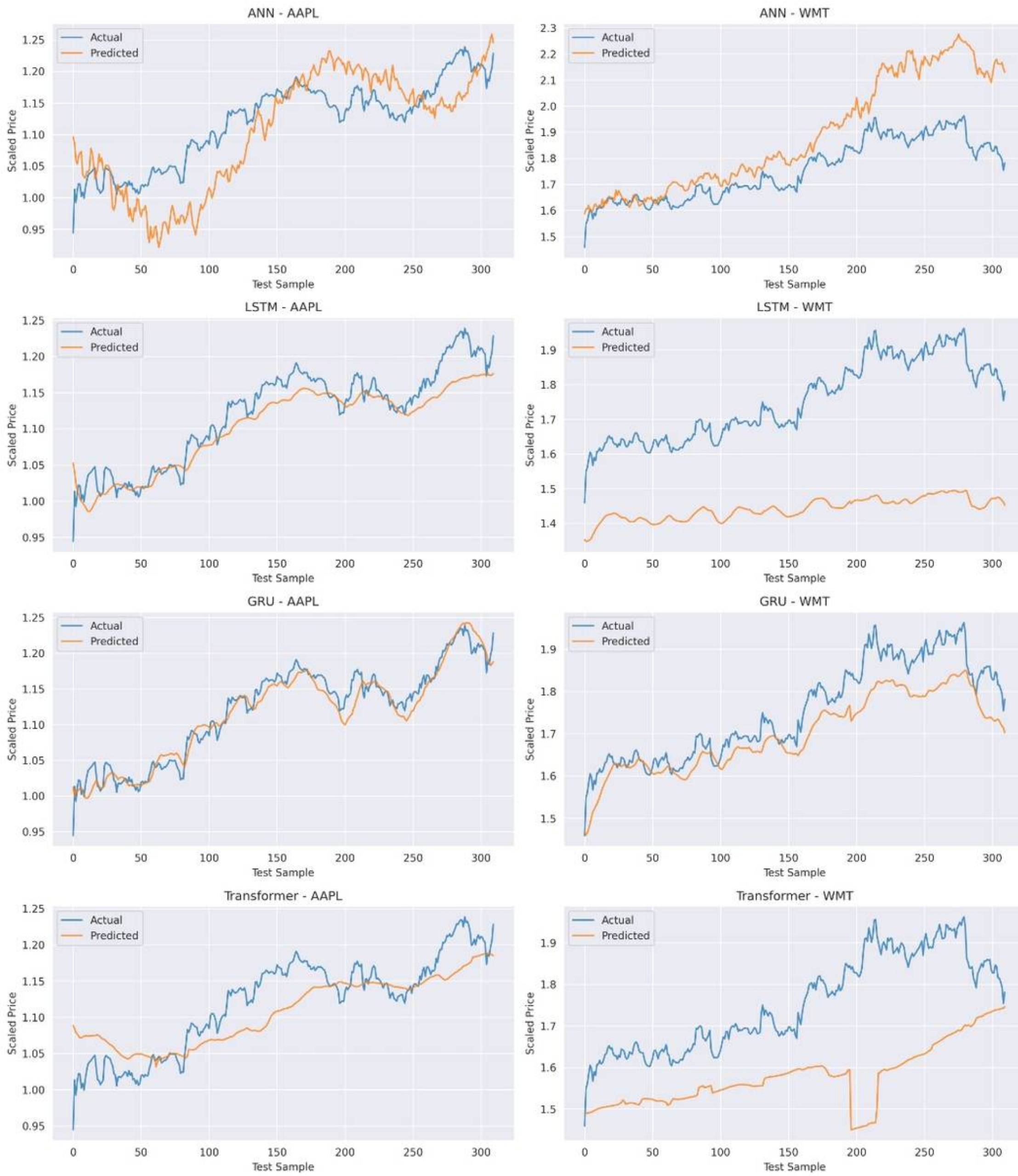

Figure 9: Predicted versus actual price traces across all four models and five stocks.

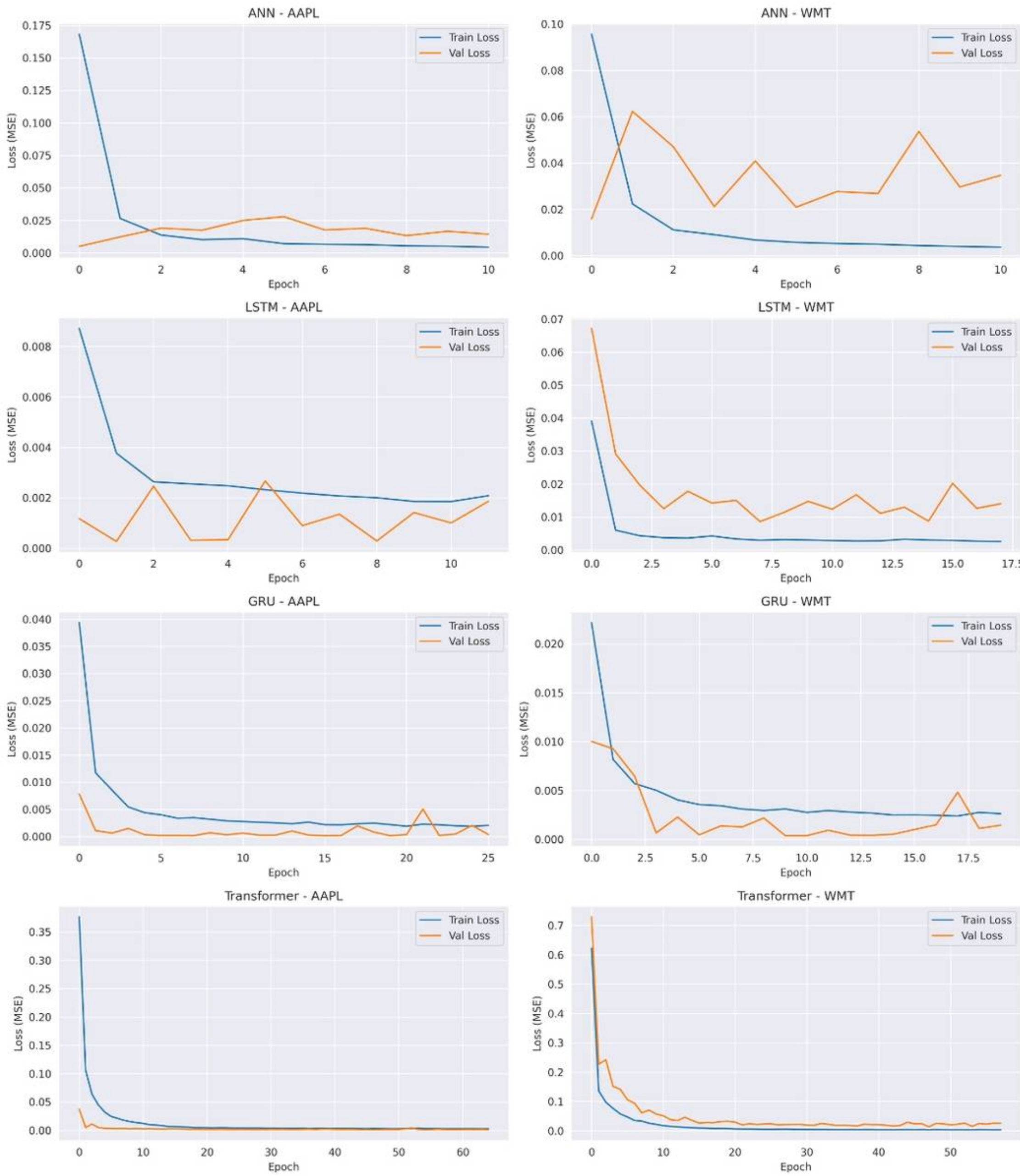

Figure 10: Training and validation loss curves by epoch, per model.

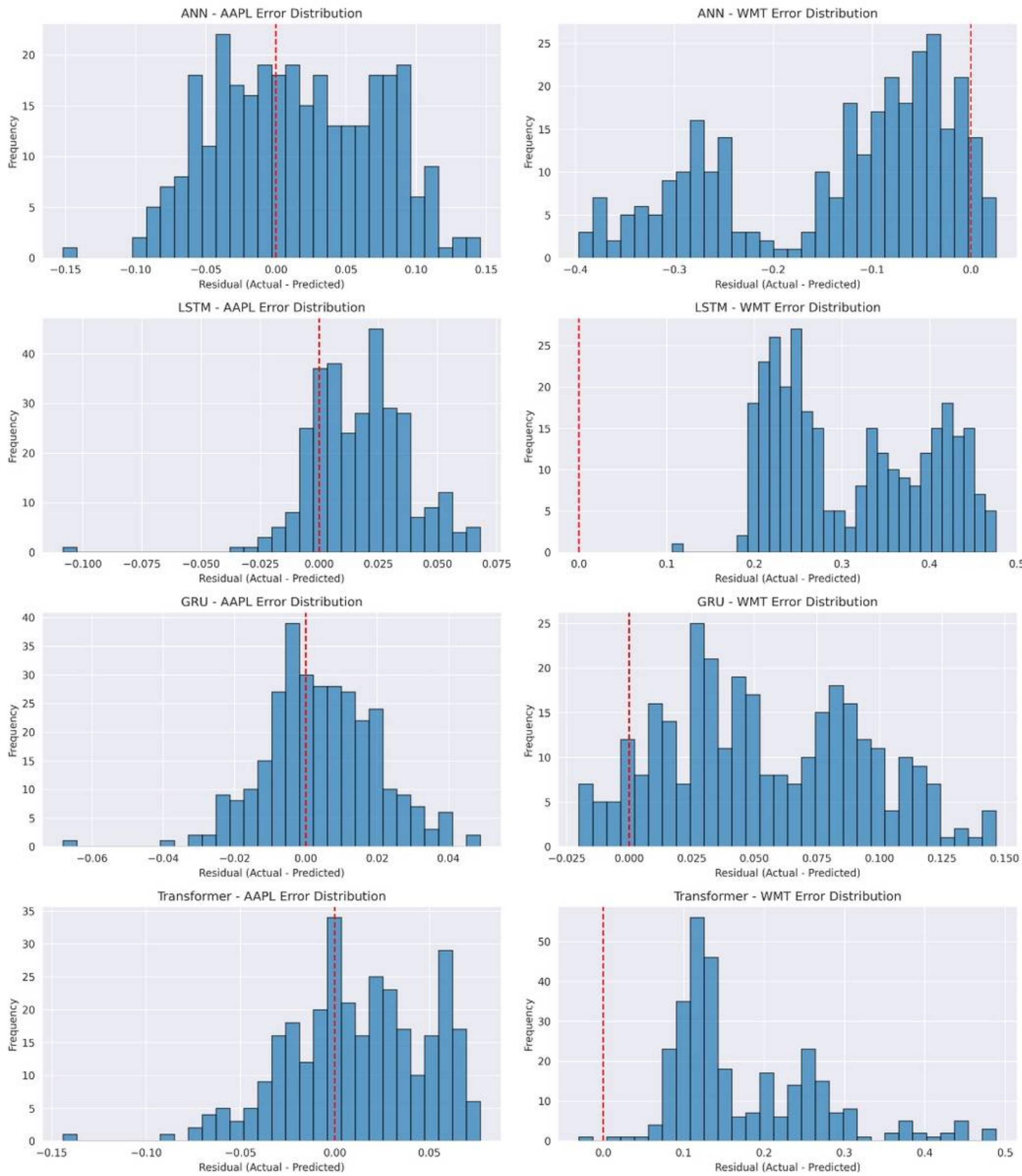

Figure 11: Distribution of prediction residuals (errors) per model.

6 Comparative Analysis

6.1 Which Model Performed Best

GRU was the best-performing architecture overall, winning outright on four of five stocks (AAPL, JPM, XOM, and WMT) and achieving the highest average R^2 of 0.4189. LSTM took the top spot only on ABT (R^2 = 0.9713), showing GRU's dominance was near-universal across this dataset rather than an average-case win alone.

6.2 Which Model Overfit

The Transformer displayed the clearest overfitting signature: despite training for the longest average duration (59.6 epochs, including a full 100 epochs on XOM), it still finished with the second-worst average R^2 (-1.4397). This combination of long training time and weak generalization is a classic overfitting pattern, likely driven by the Transformer's higher parameter count relative to the limited training data available (roughly 1,700 sequences per stock).

6.3 Which Model Trained Fastest

ANN trained fastest, requiring only 11.6 average epochs before early stopping triggered, owing to its lack of recurrent or attention computation. This speed did not translate into predictive accuracy, as ANN's average R^2 (-0.4667) remained negative overall.

6.4 Which Model Generalized Best

GRU generalized best, evidenced both by its consistently strong or least-negative R^2 across every stock and by its walk-forward validation performance on AAPL, where four of five sequential retraining folds scored R^2 between 0.65 and 0.95. Only fold five dipped to -0.36, likely reflecting a temporarily volatile period. This consistency across both stocks and time-based folds is a stronger generalization signal than any single train/test split alone.

6.5 Strengths and Weaknesses by Model

Table 4: Strengths and weaknesses summary by architecture.

Model	**Strengths**	**Weaknesses**
ANN	Fastest to train (11.6 epochs avg)	Ignores temporal order; negative R^2 on 2 of 5 stocks
LSTM	Best single result (ABT, R^2 = 0.97)	Worst average R^2 (-1.79); severely negative on JPM and WMT
GRU	Best on 4 of 5 stocks; most consistent generalizer; fastest convergence among sequence models	Still negative R^2 on WMT (-1.09), though least negative of all models there
Transformer	Captures long-range dependencies in theory	Requires the most training time yet still underperforms on 4 of 5 stocks

6.6 The WMT and JPM Difficulty Pattern

WMT and JPM were the two hardest stocks across every architecture. WMT produced negative R^2 for all four models (best: GRU at -1.09), while JPM produced negative R^2 for three of four models (only GRU remained positive, at 0.30). This pattern is consistent with a data-regime-shift explanation rather than a pure architecture failure, since the machine learning baseline comparison (Section 7.2) shows traditional models such as Random Forest also collapsed severely on both stocks — suggesting the underlying price behavior itself shifted in ways not well represented in the training data.

7 Bonus Work

7.1 Attention-Augmented LSTM

An LSTM enhanced with a multi-head attention layer was built and tested on AAPL, achieving $R^2 = 0.7334$ and MAPE = 2.53%. This is slightly lower than the plain LSTM's validated AAPL result ($R^2 = 0.7823$), suggesting that for this dataset size, the added attention mechanism did not improve upon the base LSTM — though it remains a valid and correctly implemented architectural extension worth reporting.

7.2 Machine Learning Baseline Comparison

Random Forest and XGBoost were evaluated against the deep learning models as traditional machine learning baselines. Both underperformed deep learning substantially on most stocks (e.g., AAPL: Random Forest $R^2 = -3.71$ versus GRU's 0.945), validating the choice of neural sequence models for this forecasting task. ABT was a notable exception, where Random Forest ($R^2 = 0.868$) approached LSTM's performance ($R^2 = 0.971$).

7.3 Walk-Forward Validation

Walk-forward validation on AAPL using GRU across five expanding-window folds showed strong performance in four of five folds (R^2 between 0.65 and 0.95), with one weak fold ($R^2 = -0.36$). This demonstrates that GRU's strong single-split performance mostly holds up under a more realistic, sequential retraining scenario that simulates live trading conditions.

7.4 Ten-Day Forecasting and Interactive Dashboard

Ten-day-ahead forecasts were generated for all five stocks using trained GRU models. An interactive Streamlit dashboard was also built to visualize model comparisons, forecasts, walk-forward validation results, and hyperparameter tuning outcomes within a single unified interface, supporting exploratory analysis by non-technical stakeholders.

8 Conclusion

Across five stocks and four architectures, GRU emerged as the clear overall winner, taking the top spot on four of five stocks and posting the best average R^2 (0.4189) with the most efficient training among sequence-based models. The Transformer underperformed relative to its training cost, and LSTM's strength was narrow, excelling only on ABT while collapsing on JPM and WMT. WMT and JPM stood out as broadly difficult stocks across every architecture and even traditional machine learning baselines, indicating that forecasting difficulty in this project was driven as much by underlying data regime shifts in those specific stocks as by model architecture choice. Future work could explore ensemble methods combining GRU and LSTM predictions, incorporation of external features such as news sentiment, and extended attention-based architectures trained on larger datasets to better realize their theoretical advantages.